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Abstract

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1. Introduction

For missions to explore distant small astroids(NASA's OSIRIS-REx[1], ESA's Rosetta[17], JAXA's Hayabusa[6, 8], etc), reliable three-dimensional models is essential throughout multiple scientific tasks. Stereophotoclinometry (SPC)[2] is an established approach to extract information from captured obtical images and thus derive models of astroids. Its practical use across planetary missions demonstrates its value of extrating data in roughly unknow environments with limited informations.

However, the image collections of real missions for SPC reconstruction are massive. Rosetta's SHAP5 model of comet 67P incorporated more than 5500 narrow-angle camera images acquired through June 2015 [16]. For OSIRIS-REx, successive Bennu models v10, v20, and v42 used 2351, 5114, and 15,325 images[11]. To obtain reliable SPC inputs, observation plan incorporated repeat passes over target astroid with overlapping imaging sequences, creating substantial redundancy. This redundancy provides margin for operation teams, but massive images with similar observation geometries can yield diminishing improvements while continuing to consume observation time, image storage, downlink capacity, and human processing effort. A planning simulations in commet 67P scenarioal so indicates avoidable over-observation by 43%, while improving the geometric mapping score [15]. Therefore, a well-planned observation strategy is essential to ensure that the images acquired for SPC reconstruction are both scientifically valuable and operationally efficient.

Existing works generate observing plan of astroids by optimizing the SPC-inspired observation geometry and coverage scores[14], which later be used in deep reinforcement learning to coordinate orbit selection and imaging [3]. Piccinin et al. [15] learned when to acquire images along a prescribed trajectory, reducing image count while improving geometric mapping scores and transferring the policy across targets. Wang et al. [18] combined a genetic algorithm with local search to reduce observing positions and images under illumination, visibility, and coverage constraints. However, while SPC is an algorithm which generates and refines models round after round, these planners don't take regional quality diagnostics of a newly reconstructed SPC model from last round into consideration, and their reported evaluations don't bennefit the final SPC model, they omit the modeling process and its feedback.

SPC validation and sensitivity studies have assessed the geometric accuracy of reconstructed models and the contributions of their input images. Al Asad et al. [1] developed a validation framework for Bennu shape models that combined stereo residuals, height dispersion among overlapping maplets, comparisons between rendered and observed images, and independent laser-altimeter measurements, enabling model quality to be assessed beyond internal consistency. Weirich et al. [19] rendered images of a synthetic asteroid with known topography, reconstructed its global shape using SPC, and compared the reconstruction with the truth model after registration to calibrate internal quality metrics against measured 3D error. Palmer et al. [12] varied image sets and processing conditions in synthetic-terrain experiments, combining terrain differences with normalized cross-correlation between simulated observations and images rendered from reconstructed models to evaluate geometric error and reveal deficiencies in fine surface detail. Focusing on the role of imaging conditions, Palmer et al. [13] compared SPC reconstructions from prescribed combinations of viewing and illumination geometries and image counts, identifying a compact set of four complementary topographic images and one albedo image near local noon that produced high-quality terrain in their tests. These studies provide quantitative methods for assessing reconstruction accuracy and evidence of how image combinations support

terrain and albedo recovery. Although their findings can inform observation design, they do not provide an planner that bennefits their findings.

To address these limitations, this paper propose a deep reinforcement learning framework for SPC observation planning. First, this paper theoretically describe reconstruction quality through height uncertainty and how the captured images interacts with it. It identify poorly constrained regions and provide a spatial description of the remaining reconstruction needs. Second, we establish an uncertainty critic and a reconstruction reward within the reinforcement learning framework. The planner proposes camera poses from the current surface geometry and uncertainty derived from SPC reconstruction last round. We define the reward as the aggregated reduction in surface uncertainty after the new images are incorporated in reconstruction. This realized gain trains the critic to predict reconstruction improvements, which then guides planner updates toward observations with greater expected uncertainty gains. Finally, this paper evaluate imaging efficiency using the number of acquired images and assess the geometric accuracy of the final SPC models. Compared with manually planned observation campaigns, the proposed method achieves comparable reconstruction accuracy while acquiring fewer images, thereby conserving imaging resources. Compared with existing automated SPC observation planners evaluated primarily through observation geometry and coverage scores, the proposed framework delivers final SPC surface models as data products and quantitatively assesses their geometric accuracy, enabling planning performance to be evaluated in terms of both imaging efficiency and the quality of the resulting models.

2. Related Work

SPC reconstructs surface topography and relative albedo by combining stereo geometry with image shading. Gaskell et al. [9] demonstrated its use for small-body characterization and navigation, representing local terrain with maplets and combining these patches into global shape models. Gaskell et al. [10] formalized the coupled estimation of maplet topography, landmark positions, and spacecraft position and pointing, using stereo, limb, and overlapping-maplet constraints to anchor elevations recovered from photometric slopes. Barnouin et al. [2] and Palmer et al. [11] described mission workflows that register images to an initial shape, iteratively refine local terrain, and introduce finer maplets as higher-resolution imagery becomes available. More recently, Driver et al. [7] introduced Photoclinometry-from-Motion (PhoMo), incorporating photoclinometric constraints into a keypoint-based structure-from-motion framework. Its factor graph jointly estimates camera poses, landmark positions, Sun-relative directions, surface normals, and albedos, reducing reliance on manual maplet alignment.

Studies of SPC image effectiveness examine how viewing geometry, illumination, and input errors affect the accuracy of reconstructed terrain. Al Asad et al. [1] validated successive Bennu shape models using stereo residuals, height dispersion among overlapping maplets, comparisons between observed and rendered images, and independent laser-altimeter measurements. Weirich et al. [19] reconstructed a synthetic asteroid from simulated images and compared internal SPC quality indicators with three-dimensional errors relative to the known surface, providing a controlled assessment of what those indicators reveal about geometric accuracy. Palmer et al. [12] varied input images and processing conditions in synthetic-terrain experiments, evaluating terrain differences and image correlation to identify errors in fine-scale relief and near steep slopes. They also showed that an incorrect photometric model can distort the vertical scale of local relief without being exposed by internal consistency or image-correlation checks. Focusing on observation geometry, Palmer et al. [13] tested combinations of viewing directions, illumination conditions, and image counts. Four complementary topographic images bracketing the target from north, south, east, and west, together with one albedo image near local noon, produced high-quality digital terrain models in their experiments. Replicating images without adding new observing geometries yielded little improvement, demonstrating the importance of complementary constraints within an image set; the tested five-image configuration applies to local terrain under the evaluated conditions.

Observation planning for SPC translates reconstruction requirements into choices of observing orbits, camera pointing, and image acquisition. Pesce et al. [14] formulated autonomous small-body navigation and mapping as a partially observable decision problem and selected sequences of observing orbits through a reduced space of orbital macroactions. Their reward accumulates surface-facet scores derived from emission and incidence angles, solar and spacecraft azimuths, and observation history, favoring coverage with geometries suited to SPC. Wang et al. [18] used a low-resolution shape model to formulate joint selection of observing positions and imaging actions under illumination, visibility, and repeated-observation constraints. They combined a genetic algorithm with local search to reduce the numbers of positions and images while maintaining geometrically effective coverage, and incorporated Monte Carlo sampling of position and pointing perturbations to improve plan robustness. Complementing acquisition planning, Zhang et al. [20] used KD-tree searches to identify overlapping images in unordered asteroid archives and select inputs satisfying SPC observing constraints. Their local Vesta example reduced 70 candidates to 22 overlapping images and then to five images with complementary geometries, followed by SPC reconstruction and comparison with a reference terrain model. This selection operates after acquisition, whereas the orbit and imaging planners optimize geometric observation

criteria without using regional errors from successive SPC reconstructions to choose subsequent observations.

Reinforcement learning has been applied to observation planning in small-body mapping, active three-dimensional reconstruction, and Earth observation. Chan and Aghamohammadi [3] combined deep reinforcement learning with navigation and mapping models to coordinate spacecraft maneuvers, imaging, and downlink under onboard memory and operational constraints. Piccinin et al. [15] compared Neural Fitted Q and Deep Q Network policies for deciding whether to acquire an image along prescribed spacecraft motion and pointing. Their policies use observing geometry, previous observations, mapping statistics, and remaining storage, and were evaluated on Eros, Itokawa, Bennu, and comet 67P. In one 67P simulation, the learned policy acquired 285 images instead of 500 under uniform scheduling while improving the geometric mapping index. For active reconstruction, Chen et al. [4] proposed GenNBV, which learns a next-best-view policy over continuous camera positions and orientations using proximal policy optimization (PPO). The policy combines geometric occupancy, image features, and pose history, and rewards increases in surface coverage to support generalization to unseen objects and scenes. For agile Earth-observation satellites, Chun et al. [5] represented observation tasks and time-dependent attitude-transition constraints as a graph and trained a graph-attention decision network with PPO to construct profitable task sequences with short computation times. These applications motivate learned sequential observation policies, while our focus is to connect observation decisions to regional quality changes measured after SPC updates and evaluate image savings against final geometric accuracy.

3. Method

4. Experiments

4.1. Comparative Experiments

4.2. Ablation Studies

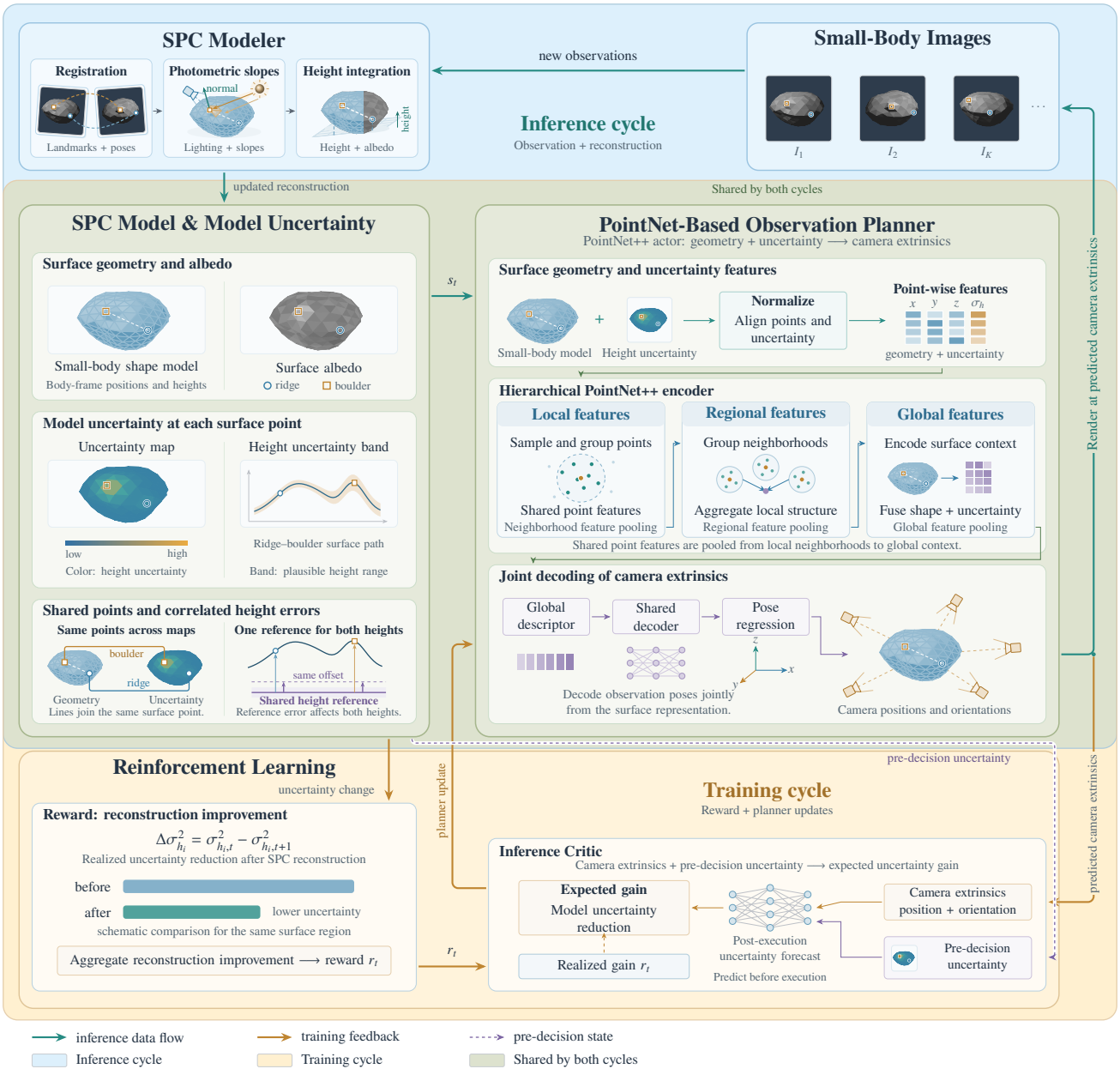


Figure 1. Overview of observation planning for an irregular small body. Blue and amber backgrounds mark the inference and training cycles; their green overlap contains the shared model state and planner. The SPC data block provides the current model and uncertainty state. Colored lines connect identical surface points across geometry and uncertainty views. Both height measurements start from one shared reference; an offset of this reference affects both heights, illustrating correlated errors. The PointNet++ planner predicts camera extrinsics from geometry and uncertainty. The inference critic uses these extrinsics and pre-decision uncertainty to estimate the expected uncertainty reduction. Realized gains train the critic, which guides planner updates. Teal and ochre arrows denote inference data flow and training feedback; the dashed purple connection carries pre-decision uncertainty.

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